

The efficiency of simulated annealing around T_c has been reported in numerous studies. Kirkpatrick et al. (1983) demonstrated that during phase transition, the search becomes more efficient. Strobl & Barker (2016) showed that when using simulated annealing to solve phylogeny reconstruction, the search is constrained to a small valley of the search space when the temperature is below T_c . The search efficiency around T_c can also be seen in Figure 2 of (Cai & Ma, 2010b).

While some algorithms (Basu & Frazer, 1990; Cai & Ma, 2010a) have used critical temperature T_c to design the initial and final temperatures, its use for reheating is less common. Abramson et al. (1999) proposes a reheating strategy tied to function cost that implicitly involves T_c . However, this method lacks theoretical support and is sensitive to hyperparameters. Instead, our approach directly reheats to the critical temperature, which injects just the right amount of stochastic energy back into the system, enabling it to escape “wandering in contours” without incurring excessive randomness of a high-temperature regime.

5.2.2 Specific Heat

Determining the critical temperature in simulated annealing requires identifying the phase transition point during optimization, which is characterized by peaks in *specific heat* (Kirkpatrick et al., 1983; Strobl & Barker, 2016). In statistical physics, the system’s energy at temperature T , denoted as $E(T)$, adheres to the Boltzmann distribution. Thus, the expected energy of a system can be seen as a function of the temperature, $\mathbb{E}[E(T)]$. The specific heat at temperature T , denoted as C_T , is traditionally defined in thermodynamics as the rate of change of the expected energy $\mathbb{E}[E(T)]$ to temperature (Aarts et al., 1987), given by $C_T = \frac{\partial \mathbb{E}[E(T)]}{\partial T}$. Integrating over the Boltzmann distribution allows deriving specific heat in terms of energy variance:

$$C_T = \frac{\sigma^2(E(T))}{T^2} \quad (9)$$

where $\sigma^2(E(T))$ is the variance of the system energy at T .

5.2.3 Determination of Critical Temperature

To determine the critical temperature based on specific heat, we first denote $T(t)$ as the temperature at step t , $C(t)$ as the specific heat at temperature $T(t)$, and x_t as the solution sampled at step t , then by selecting an appropriate sample size M , we define the approximation of $C(t)$ as :

$$\hat{C}(t) = \frac{\sigma^2(\{f(x_{t-M+1}), \dots, f(x_t)\})}{T(t)^2}, \quad t \geq M \quad (10)$$

where $\sigma^2(\{f(x_{t-M+1}), \dots, f(x_t)\})$ represents the variance in objective values over the M most recent steps. The critical temperature T_c can be determined as $T(t^*)$, where $t^* = \arg \max_{t \geq M} \hat{C}(t)$. However, SA with gradient-based discrete samplers converges rapidly in the initial stage, resulting in an *abnormal* initial peak in specific heat (due to the high variance). As the annealing progresses, the specific heat quickly decreases, eventually stabilizing at a level more typical of a critical temperature. This behavior is shown in Figure 4a&4b.

To address the abnormal peak, we introduce a “skip step” threshold, denoted as t_{skip} , ensuring that the initial transient behavior is excluded from the analysis. Thus, the critical temperature is identified as $T(\tilde{t}^*)$, where

$$\tilde{t}^* = \arg \max_{t \geq t_{\text{skip}}} \hat{C}(t). \quad (11)$$

Our method diverges from traditional SA reheat strategies (Abramson et al., 1999) by accounting for inhomogeneous chains and addressing gradient-based methods’ unique abnormal peaks.

5.3 Reheated Gradient-based Discrete Sampling for Combinatorial Optimization

Combining results in Section 5.1 and 5.2, we obtain the *Reheated Sampling for Combinatorial Optimization* algorithm (ReSCO), which is summarized in Algorithm 1. ReSCO is compatible with any gradient-based